# Back squat · 40 kg · 23 %

Bertrand Cadet · 27 Oct 2026 at 17:07 · Made

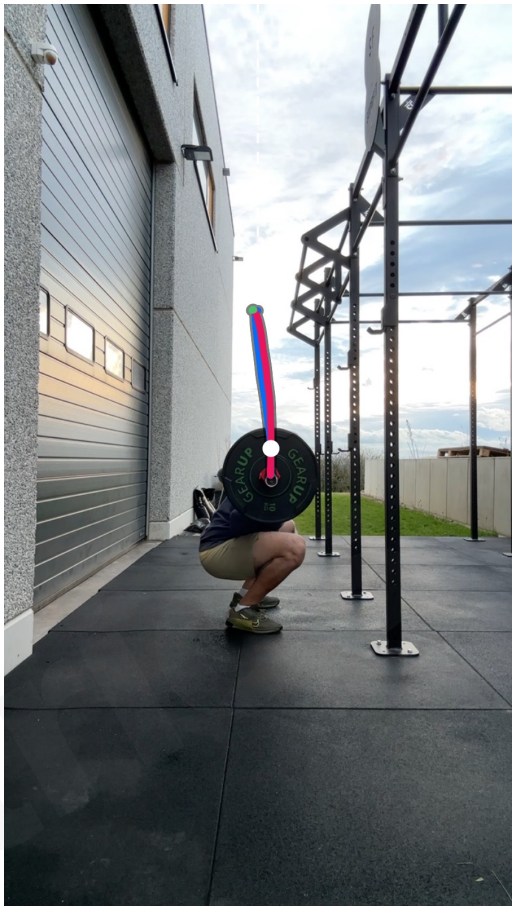
**92** / 100 · technique score (bar path)

| Criterion | Points |
|---|---|
| Vertical bar path | 37 / 45 |
| Velocity loss | 35 / 35 |
| Bar tilt | 20 / 20 |

## Fix first

- **Horizontal bar drift: decrease**
  6,0 cm for a goal of ≤ 5 cm: reduce by 1,0 cm — keep a more vertical bar path: bar over mid-foot in the squat and deadlift, a consistent J-curve in the bench press.

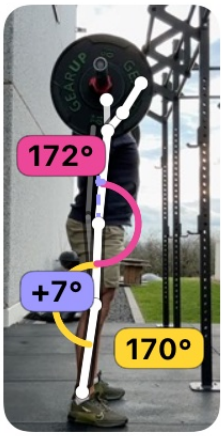
## Bar path



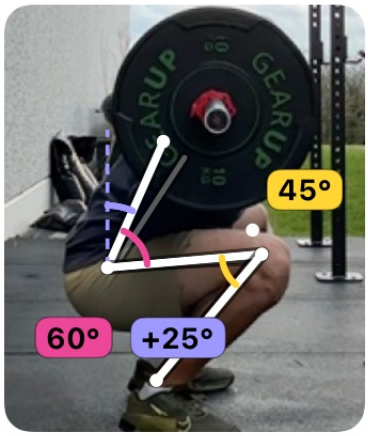
| Velocity (sensor) | | |
|---|---|---|
| **Measurement** | **Value** | **Goal · to do** |
| Mean ascent velocity<br>key metric of velocity-based training | 0,65 m/s | |
| Peak ascent velocity | 0,58 m/s | |
| Velocity loss in the set<br>relative to the fastest rep (0,72 m/s) | 10 % | goal ≤ 20 %<br>OK |
| Mean descent velocity | 0,41 m/s | |
| Descent time | 1,98 s | |
| Ascent time | 1,17 s | |
| Mean ascent power | 164 W | |
| Peak power<br>2,2 W/kg of body mass | 246 W | |

| Bar path | | |
|---|---|---|
| Range of motion<br>vertical bar travel | 76 cm | |
| Horizontal bar drift<br>max drift from the starting position | 6,0 cm | goal ≤ 5 cm<br>↓ reduce by 1,0 cm |
| Position at the top<br>relative to the starting position | -1,8 cm | |
| Max tilt<br>one side higher than the other | 0,4° | goal ≤ 3 °<br>OK |
| Max horizontal rotation<br>one side is ahead of the other | 0,4° | |

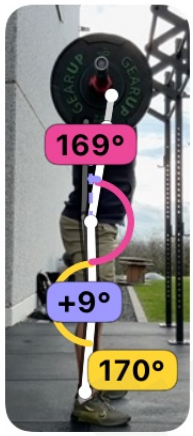
## Posture: positions at key moments




Start




Bottom




Lockout

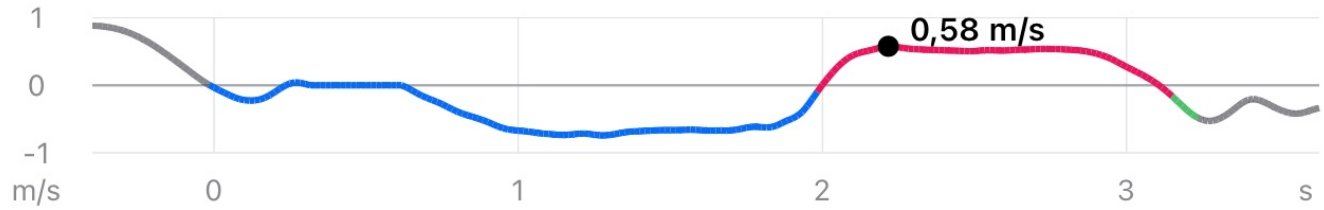
**Posture (angles read from the video)**

| | Knee | Hip | Trunk | Elbow |
|---|---|---|---|---|
| Start | 170° | 172° | +7° | — |
| Bottom | 45° | 60° | +25° | 155° |
| Lockout | 170° | 169° | +9° | — |

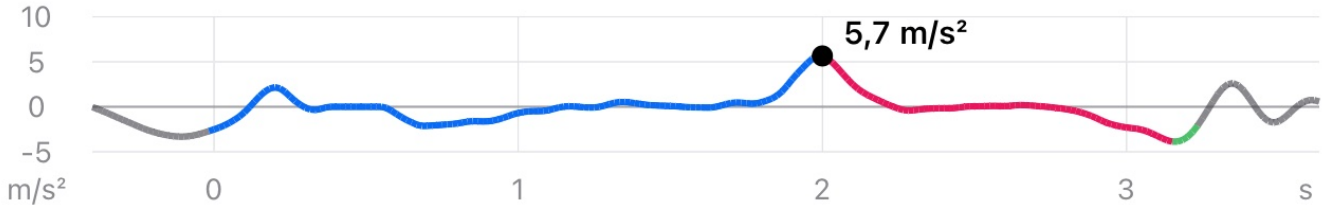
Depth: hip below the knee at the bottom

Drawings: yellow = knee, pink = hip, purple = trunk relative to vertical, green = elbow (when bent). Knee, hip and elbow 180° = straight; trunk 0° = vertical, + leaning forward, − leaning back. 2D side-on measurement, accurate to a few degrees.

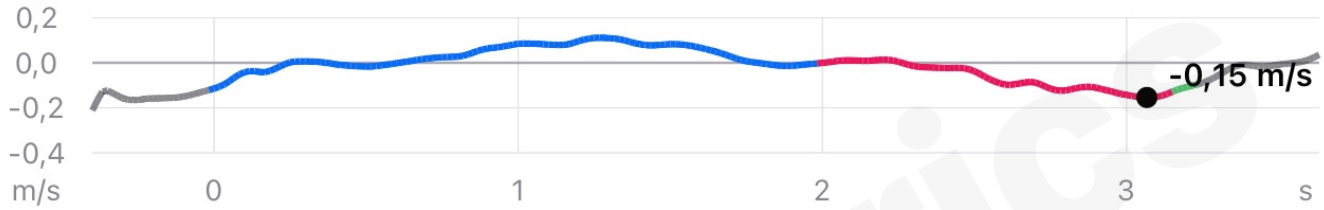
## Velocity / time (m/s)



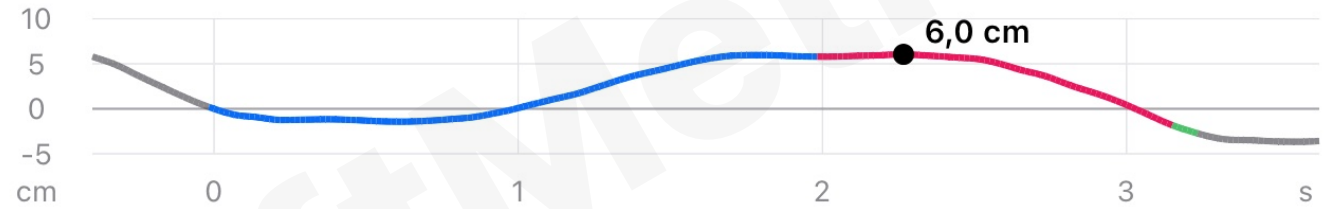

## Acceleration / time (m/s²)




## Horizontal velocity / time (m/s)




## Horizontal displacement / time (cm)




## Power / time (W)

smoothed video curve; peak measured by the sensor: 246 W




LiftMetrics

# How to read these numbers

| | |
|---|---|
| **Goal · to do** | Target range for each measurement (adjustable in Settings › Goals). Green = within range · orange = outside but within tolerance · red = beyond. “increase / decrease by X” = gap to close to get back in range. |
| **Score /100 (strength)** | Vertical bar path 45 points, velocity loss 35, bar tilt 20. A criterion that is not measured is left out of the calculation. |
| **Green / orange / red colors** | Green = on target · orange = slightly outside (within tolerance) · red = to correct. The ranges for each measurement are given below. |
| **Bar path colours (strength)** | Blue = descent · pink = ascent · yellow = sticking point · green = lockout. Velocity chart: same colours. |
| **Mean V (mean concentric velocity)** | Vertical distance covered during the ascent divided by its duration (m/s). It is the reference measure of velocity-based training: the heavier the load, the lower it gets. |
| **Peak V (strength)** | Peak vertical velocity during the ascent (m/s). |
| **Loss** | Mean velocity loss compared with the fastest rep of the set (ref.). Shows fatigue within the set. goal ≤ 20 %, orange up to 10 % outside, red beyond. |
| **Range of motion** | Vertical distance travelled by the bar between the lowest and highest point (cm). |
| **Drift** | Largest horizontal gap between the bar and its starting position (cm): a truly vertical path gives a small value. Squat: goal ≤ 5 cm, orange up to 3 cm outside, red beyond. |
| **Ascent** | Duration of the ascent (s). |
| **Power** | Mean power of the ascent (W), computed with the load. |
| **Tilt** | Maximum bar tilt measured by the sensor (one side higher than the other), in degrees. Squat: goal ≤ 3 °, orange up to 2 ° outside, red beyond. |

LiftMetrics